



Assignment 1

1701ICT— Creative Coding

School of ICT

Griffith University

Trimester 1, 2024

Milestone 1 (20%) is due at the end of week 3 with demonstrations to be held in your week 4 workshop. **(11:59pm Sunday 24th March)**

Milestone 2 (20%) is due at the end of Week 6 with demonstrations to be held in your week 7 workshop. **(11:59pm Sunday 21st April)**

Instructions

- **Due:** Milestone 1, 11:59pm Sunday 24th March. Milestone 2 11:59pm Sunday 21st April)
- **Marks:** 40% of your overall grade (20% each milestone)
- **Late Submissions:** Late submission is allowed but a penalty applies. The penalty is defined as the reduction of the mark allocated to the assessment item by 5% of the total weighted mark for the assessment item, for each day that the item is late. Assessment items submitted more than five working days after the due date will be awarded zero marks.
- **Extensions:** You can request for an extension of time on one of two grounds;
 - medical
 - other (e.g., family or personal circumstances, employment-related circumstances, unavoidable commitments).All requests must be made through the myGriffith portal.
- **Individual Work:** You must complete this assignment individually.
- **Presentation:** You **must** present/demonstrate your work in your week 4 workshop (milestone 1) and week 7 workshop (milestone 2) to a teaching team member. Your work **will not be marked** if you do not present it.
- **Other:** This assignment is a mandatory pass component. You must attain 40% (16/40) to be eligible to pass the course.

Overview

The purpose of the assignment is to assess your ability to implement simple interactions, shapes and objects using JavaScript and p5.js. Each Question should be submitted on L@G in a separate .js file (titled q1.js, q2.js etc.), but you do not need to submit html and css files. Some of the questions will require you to investigate some concepts outside of the lecture notes. ***Please make sure that all code that you submit is your own and is not taken/copied from anywhere else.***

Milestone 1 (due 11:59pm Sunday 24th March)

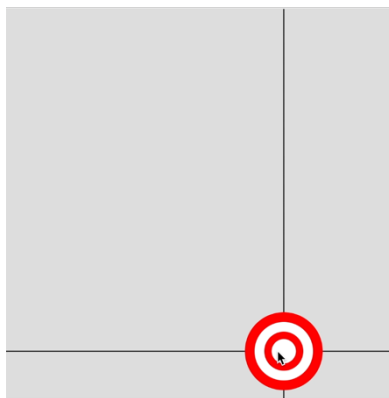
Important! For this milestone, you can only use the p5js functions covered on weeks 1-3.

Question 1 (4 marks)

For this question, draw a bullseye target that follows the user's mouse around the screen (hint: mouseX, mouseY). The target should have at least 3 circles inside of each other and look similar to:



In addition, draw crosshairs (horizontal and vertical lines) that move together with the target as shown in the video provided below:



Question 2 (4 marks)

Using the following data and p5js to draw a pie chart about favorite types of movie.

Type	Count
Comedy	4
Action	5
Romance	6
Drama	1
SciFi	4

You should annotate each slice of the pie chart with its label, value and percentage (either label the pie chart directly, or do a legend).

Your code should allow for different pie charts to be created by storing the count for each movie type in variables that can be changed.

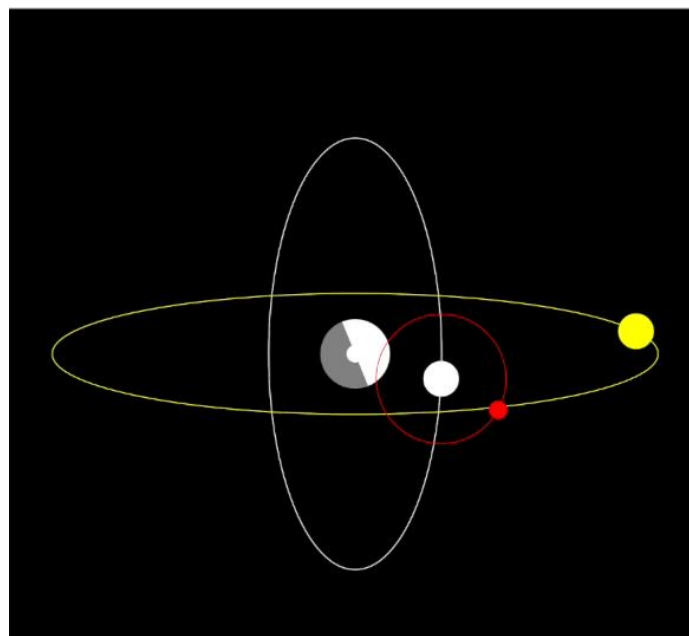
Question 3 (12 marks)

For this question, you will draw an animated planet and moons using p5js.

- a. Set the background colour to black. Draw the planet as 2 half circles, one bright, one dark.
Draw two smaller circles to represent the moons, one yellow and one white. The size of each moon should be half the size of the planet. Position the moons using the angular arithmetic used in the last slide of 3.4 Math functions.
Animate the angle of the two moons at **different speeds** so that the moons will orbit the planet on an **elliptic** track, one faster than the other. Draw the track of each moon in a similar colour to its respective moon. (4 marks)
- b. Add an additional smaller moon that orbits one of the other moons. (This will mean you should have 3 moons total now). (2 marks)
- c. Have which half of the planet is dark or light change based the position of one of the larger moons (either the yellow or white). This means the planet will spin such that its bright half will always face the selected moon. (2 marks)
- d. Add a small circle in the centre of the planet. Colour this circle based on which moon is closest, i.e. white if the white moon closest, yellow if the yellow closest. (2 marks)

Important Notes!

For all parts of this question, it must be easy to change the size and position of the planet in the code. The moons' size and position should be automatically updated accordingly. (2 marks)



Milestone 2 (due end of Week 6, 11:59pm Sunday 21st April)

Question 1 (8 marks)

Sol LeWitt is a famous American artist. He came to fame in the late 1960s with his wall drawings and "structures" (a term he preferred instead of "sculptures") but was prolific in a wide range of media including drawing, printmaking, photography, painting, installation, and artist's books (https://en.wikipedia.org/wiki/Sol_LeWitt). In this exercise, you are asked to recreate the works of Sol LeWitt using p5js as per the below instructions.

- a. Six-part drawing. The wall is divided horizontally and vertically into six equal parts. 1st part: On red, blue horizontal parallel lines, and in the center, a circle within which are yellow vertical parallel lines; 2nd part: On yellow, red horizontal parallel lines, and in the center, a square within which are blue vertical parallel lines; 3rd part: On blue, yellow horizontal parallel lines, and in the center, a triangle within which are red vertical parallel lines; 4th part: On red, yellow horizontal parallel lines, and in the center, a rectangle within which are blue vertical parallel lines; 5th part: On yellow, blue horizontal parallel lines, and in the center, a trapezoid within which are red vertical parallel lines; 6th part: On blue, red horizontal parallel lines, and in the center, a parallelogram within which are yellow vertical parallel lines. The horizontal lines do not enter the figures. (6 marks)
- b. Draw a wall with "irregular wavy color bands". Hint: you should use `curveVertex` and random functions in p5js library. (2 marks)

Note: You should use loops to draw the lines, not manually write 50-100 line functions

Question 2 (6 marks)

In this exercise you will implement a game with a simple interaction with a ball.

The game starts when you click the UP arrow on the keyboard and a ball appears moving up from the bottom of the canvas. You can then change the direction of the ball using the four arrow keys. The ball will stop moving only when you click the mouse.

When the ball goes beyond the canvas it should appear back on the other side.

(A demo will be provided on the course page, under Assignment 1 instructions)

Question 3 (6 marks)

In order to attract new students, we need your helps to design a dynamic banner to promote this course and p5js. You are free to choose shapes, colors, animations ... but your implementation needs to satisfy these following constraints:

- a. Implement using p5js.
- b. Include at least 2 custom functions.
- c. Include at least 1 loop.
- d. Include animation.
- e. Harmony in design.